TO 33kV SWITCHBOARD F00-SW-010 PANEL B23
REFER 2000-F00-SLD-EL-10006

INTERTRIPS TO SIGNALS

F23-NR-016
FINAL TAILS TRANSFORMER
F23-TR-016 NEUTRAL EARTHING CUBICLE

F23-TR-016
33 /11 kV
5000 kVA
Dyn11
z=7%
RNAN

254Ω +/-5%
30/0.05A 25A FOR 10s
SENSING
ER-15kV

TRANSFORMER PROTECTION DEVICES TERMINAL BOX

F23-SR-001
FINAL TAILS PUMPS 11kV & 415V SWITCHROOM

E-SHUT
F23SR001-LCS
NOTE 5

F23-SW-016
FINAL TAILS PUMPS 11kV SWITCHBOARD

NOTE 4

TEST

300/1A
2.5VA
Cl 0.2

PQM
SEL735

PR
SE-330AU

EARTH FAULT INDICATION

TRANSFORMER INSTRUMENTATION ALARM & TRIP CONTACTS (NOTE 6)

UPSTREAM TRIPS

300/1A
2.5VA
5P20

CLOSE
TRIP
630A

NOTE 2
50 51 50G 51G AFD 50N 51N
PR
SEL 751
49 63 71

2000-F23-SCM-EL-10101
A01
NOTE 2
NOTE 3
A02
A03
A04
2000-F23-SCM-EL-10107
A05
A06
11kV 1250A 3~E 25kA FOR 3s
110V AC SUPPLY

2A
630A
NOTE 2

VT
11kV/√3 / 0.11kV/√3 / 0.11kV/√3
15VA CLASS 0.2
20VA CLASS 3P

160E
CLOSE
50 51 50G 51G 27 81 AFD
PR
SEL 751

50/1A
2.5VA
5P20

50/1A
1.5VA
10P10

VSD CONTACTOR CLOSE PERMISSIVE
VSD HEALTHY
VSD FAULT

F23PP466-VS
FINAL TAILS
PUMP 1 VSD
11/3.3kV 800kW

F23PP467-VS
FINAL TAILS
PUMP 2 VSD
11/3.3kV 800kW

F23PP468-VS
FINAL TAILS
PUMP 3 VSD
11/3.3kV 800kW

F23PP469-VS
FINAL TAILS
PUMP 4 VSD
11/3.3kV 800kW

M 3~ RTD
800kW

F23-PP-466
FINAL TAILS
PUMP 1
2000-F23-SCM-EL-10103

F23-PP-467
FINAL TAILS
PUMP 2
2000-F23-SCM-EL-10105

F23-PP-468
FINAL TAILS
PUMP 3
2000-F23-SCM-EL-10109

F23-PP-469
FINAL TAILS
PUMP 4
2000-F23-SCM-EL-10111

CONT'D ON SHEET 2 DRG 2000-F23-SLD-EL-10002

NOTES

1. THE SWITCHBOARD SHALL BE EXTENSIBLE ONE END FOR FUTURE EXPANSION AND SUPPLIED WITH ONE SPARE TIER AS SHOWN.
2. ARC FLASH IN INCOMER CIRCUIT BREAKER COMPARTMENT IS NOT CURRENT SUPERVISED.
   TRIPPING OF BUS AND DOWNSTREAM CB/FC COMPARTMENTS IS CURRENT SUPERVISED.
3. ARC FLASH POINT DETECTORS SHALL BE PLACED IN THE CABLE ZONE, CIRCUIT BREAKER CHAMBER OR FUSE CONTACTOR CHAMBER AND THE BUSBAR CHAMBER. THE DETECTORS SHALL BE CONNECTED DIRECTLY TO THE PROTECTION RELAY IN EACH PANEL.
4. ONE OFF 30/0.05A PROTECTION CT SHALL BE INSTALLED IN THE NER CUBICLE AND CONNECTED TO THE INCOMER PROTECTION RELAY TO MONITOR FOR EARTH FAULT DETECTION .
5. SWITCHROOM EMERGENCY SHUTDOWN (E-SHUT) PUSHBUTTON IS HARDWIRED DIRECT TO THE SHUNT TRIP OF THE INCOMER CIRCUIT BREAKER AND UPSTREAM 33kV FEEDER.
6. 33/11kV TRANSFORMER INSTRUMENTATION TRIP CONTACTS SHALL BE HARDWIRED TO THE 11kV SWITCHBOARD INCOMER PROTECTION RELAY WHICH SHALL INTERTRIP THE UPSTREAM 33kV FEEDER.
7. UPSTREAM 11kV FEEDER SHALL BE TRIPPED ON THE EARTH & ARC FAULT DETECTION OF THE DOWNSTREAM INCOMER ACB.

LEGEND

PR	INCOMER / FEEDER PROTECTION RELAY
PQM	POWER QUALITY METER
	ARC FLASH SENSOR (POINT TYPE) NOTES 2 & 3
	CAPACITIVE VOLTAGE INDICATION
	MECHANICAL INTERLOCK DEVICE
E-SHUT	SWITCHROOM EMERGENCY SHUTDOWN
M	MOTOR OPERATOR
27	UNDER VOLTAGE
50	INSTANTANEOUS OVER-CURRENT
50G 50N	INSTANTANEOUS EARTH/NEUTRAL OVERCURRENT
51	TIMED OVER-CURRENT
51G 51N	TIMED EARTH/NEUTRAL OVERCURRENT
59	OVER VOLTAGE
64	EARTH FAULT
49	TRANSFORMER TEMPERATURE
63	TRANSFORMER PRESSURE / BUCCHOLZ
71	TRANSFORMER LIQUID / GAS LEVEL
81	FREQUENCY
AFD	ARC FLASH DETECTION

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	
Signed [redacted]	Date 15/10/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

DRG No.	REFERENCES
2000-F23-DRG-EL-10001	F23-SR-001 11kV FINAL TAILS PUMPS SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F23-SLD-EL-10051	F23-SR-001 11kV FINAL TAILS PUMPS SWITCHBOARD KEY INTERLOCKING SINGLE LINE DIAGRAM
2000-F23-SLD-EL-10002	F23-SR-001 11kV FINAL TAILS PUMPS SWITCHBOARD SINGLE LINE DIAGRAM SHEET 2 OF 2
2000-F00-SLD-EL-10006	F00-SW-010 33kV PLANT HV DISTRIBUTION SWITCHBOARD SINGLE LINE DIAGRAM SHEET 5 OF 5
2000-F00-SLD-EL-10001	OVERALL SITEWIDE SINGLE LINE DIAGRAM
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	19SEP25	FAT REDLINES INCORPORATED					
0	22OCT24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PRIMERO PROJECT NUMBER: 23517

PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: FINAL TAILINGS HANDLING & STORAGE
F23-SW-016 11kV SWITCHBOARD
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No:	REVISION
A1	NONE	2000-F23-SLD-EL-10001	1

O:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\EL\F23 - FINAL TAILINGS HANDLING & STORAGE\2000-F23-SLD-EL-10001.DWG 02OCT25 9:04AM HOTSEAT04